

LF-VCO

This voltage-controlled low-frequency modulation oscillator also offers the option of pulse-width modulation for the square-wave signal. The frequency of the output signal is adjustable between approximately 0.005 Hz and 20 Hz.

The circuit of the LF-VCO is shown in Figure 1a. In principle, the circuit's operation is identical to that of the VC-LFOs; therefore, a detailed description is omitted here. In addition to a triangle output and two sawtooth outputs (one with a rising edge and one with a falling edge), there is also a pulse-width-modulated square-wave output.

The Pulse-Width Modulator

The pulse-width modulator (Figure 1c) is a waveform converter that is also well-suited for low-frequency applications. Housed on its own circuit board, this converter derives a square wave with an adjustable and modulatable pulse-to-pause ratio from an input sawtooth signal. The circuit (Figure 1c) consists essentially of a comparator utilizing transistors T2, T3, and T4. Varying the reference voltage produces a square-wave output signal with a variable pulse-to-pause ratio. The reference voltage is generated by combining an external modulation signal (PWM) with two adjustable components (P7, P10) via an input summer (IC4). P7 is used for the manual adjustment of the pulse width, while P10 allows for the attenuation of the external modulation signal. Trimmers P8 and P9 are used to set the modulation range (1% to 99%). An LED indicator is not strictly necessary, as the PWM frequency can be determined from the fluctuation rate of the indicator LED on the external modulation source.

Construction

As shown in Figure 2, the LF-VCO utilizes the VCLFO circuit board. Due to the additional PWM section, the LF-VCO front panel (Figure 3) accommodates only a single LF-VCO; consequently, the circuit board is not fully populated. There is space for only one LF-VCO, corresponding to the position of LFO 2 on the board. A separate circuit board is provided for the PWM section (Figure 1c); its layout and component placement plan are shown in Figure 4. This board can be mounted to the main PCB using spacers at the designated mounting holes. Figure 5 illustrates a suggested mounting arrangement. For the LF-VCO, only a low-leakage film capacitor should be used for capacitor C1. With the exception of the FREQ potentiometer, all rotary potentiometers are types with a 4 mm shaft. The circuit configuration used for IC5 is known as an impedance converter or a voltage follower. The latter term is entirely logical when one considers that the output voltage equals the difference between the voltage at the non-inverting input and the voltage between the two op-amp inputs. Since the feedback loop prevents a voltage difference from developing between the op-amp inputs, the output voltage tracks the input voltage. The term "impedance converter" is also justified because every op-amp inherently possesses high

input impedance and low output impedance, and these characteristics are not compromised by the external circuitry. The gain of this configuration is 1.

The section of the circuit board shown in Figure 6 for the pulse-width modulator illustrates the wiring for potentiometers P7 and P10.

Calibration

The oscillator section should be calibrated following the instructions for the VC-LFO module. However, the modulation range of the PWM section must also be calibrated:

Turn the wiper of P10 (on the fully wired LF-VCO module) to ground. Adjust trimmers P8 and P9 so that the entire pulse width adjustment range (1–99%) can be covered using potentiometer P5. However, if an oscilloscope is available, the following method will usually yield results faster:

- a) Set P8 so that the voltage at the wiper is -5.5 V , and set P9 to its maximum value. Then connect a multimeter (DC range) to pin 6 of IC4.
- b) Monitor the PW output signal (square wave output) on the oscilloscope screen.
- c) Set P7 for maximum and minimum pulse width (rotary potentiometer end positions) and note the voltage values on the multimeter for both positions. The higher voltage value (at minimum PW) is designated below as $U_{\max}$, and the lower value (at maximum PW) as $U_{\min}$.
- d) Now turn P8 toward ground and set P7 to the maximum wiper voltage ($+4.7\text{ V}$). Use P9 to adjust the output voltage of IC4 so that it equals the difference between the voltages measured in step c ($U_{\max} - U_{\min}$). Since IC4 is configured as an inverter, the output voltage of IC4 will also be negative.
- e) Leave P7 unchanged, then adjust P8 for maximum pulse width. Calibration is now complete. To verify, turn the wiper of P7 to ground; the output signal on the oscilloscope screen should now show minimum pulse width.

Applications

In principle, the applications of the LFVCO correspond to those of the LFOs from the FORMANT series or standard VC-LFOs. Additionally, the "PW" control allows for the setting of square waves with any desired pulse-to-pause ratio. A "phased" modulation signal can be achieved via the PWM input, enabling effects such as vibrato with varying emphasis on the upper and lower "limit frequencies."

Literature:

- M. Bertuch: VCLFO, Patchcord, No. 3, p. 16

- C. Chapman: FORMANT Book 1, Elektor Verlag

Parts list LF-VCO

Resistors (carbon layer, 5%):

R1, R15, R20 = 22k

R2, R3, R22 = 68k

R4, R5, R7, R9, R10, R16, R17, R18, R19, R21, R23 = 100k

R6 = 47k

R8, R29, R30, R34, R35 = 1k

R11, R25 = 3k9

R12, R13 = 470Ω

R14, R24, R26, R27, R31, R33 = 10k

R28 = 1k5

R32 = 4k7

R36 = 1M

Rx = 470 (see VC-LFO)

Potentiometer:

P1, P2 = 10k (trimmer)

P3 = 1k (trimmer)

P4 = 100k log. (6 mm shaft)

P5, P6, P10 = 100k (lin., 4 mm shaft)

P7 = 10k lin. (4 mm shaft)

P8, P9 = 100k (trimmer)

Capacitors:

C1 = 1 μF (MKH, MKS)

C2, C3 = 10 μF/25 V

C4 = 470 pF (MKH, MKS)

C5, C6 = 100 nF (MKH, MKS)

Semiconductors:

IC1, IC4, IC5 = μA741C (Mini-DIP)

IC2, IC3 = LM324N

T1...T3 = TUN (C-types, e.g., BC550C)

T4 = TUP (C-types, e.g., BC309C)

D1, D3, D4 = DUS

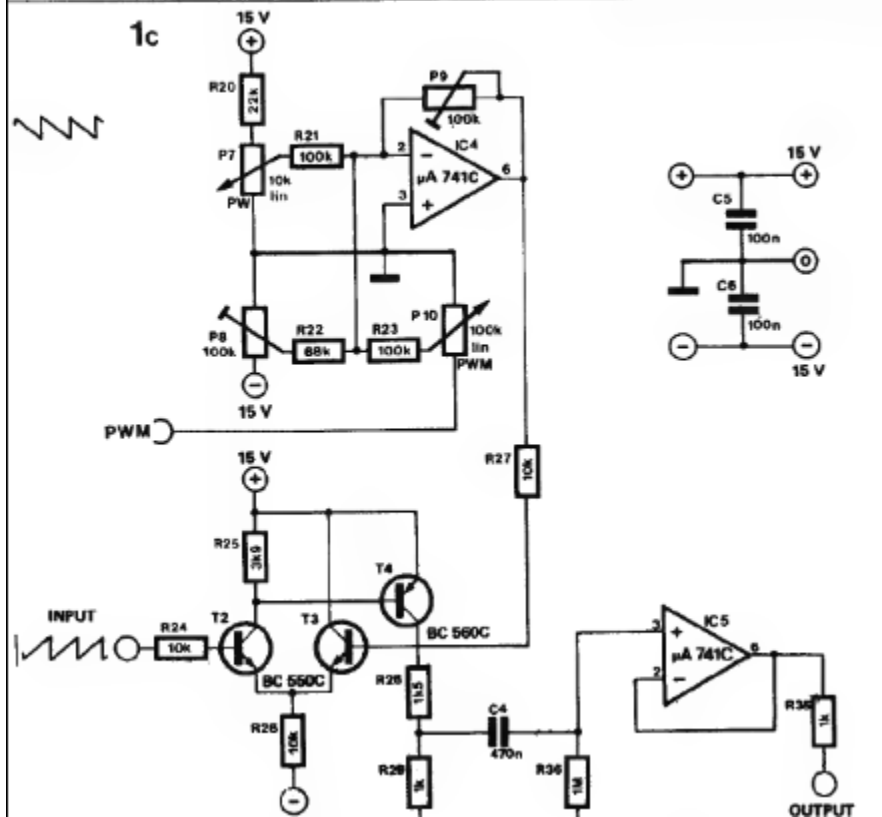
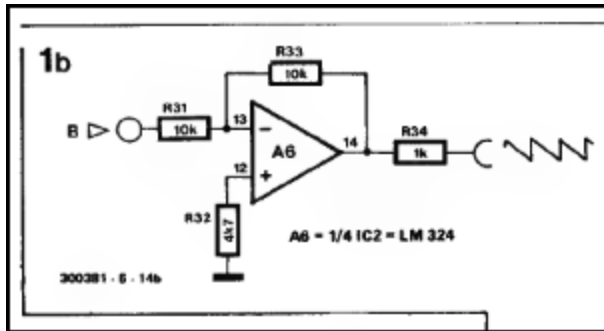
D2 = LED

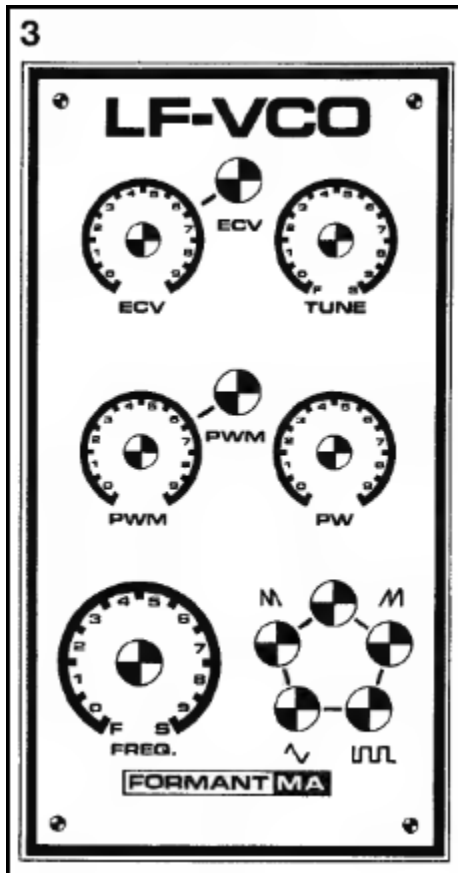
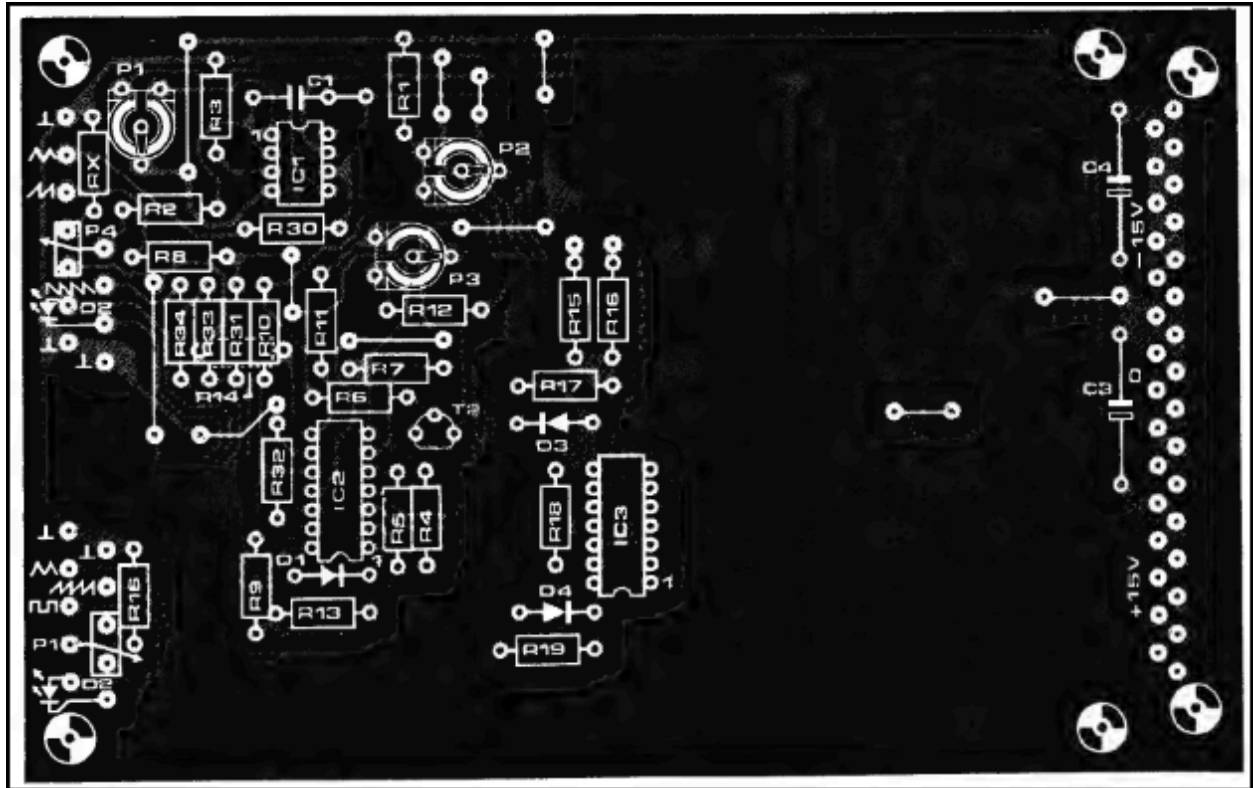
Miscellaneous:

6 x jack sockets, 3.5 mm
4 x rotary knobs (4 mm)
1 x rotary knob (6 mm)
1 x 31-pin header strip or solder pins

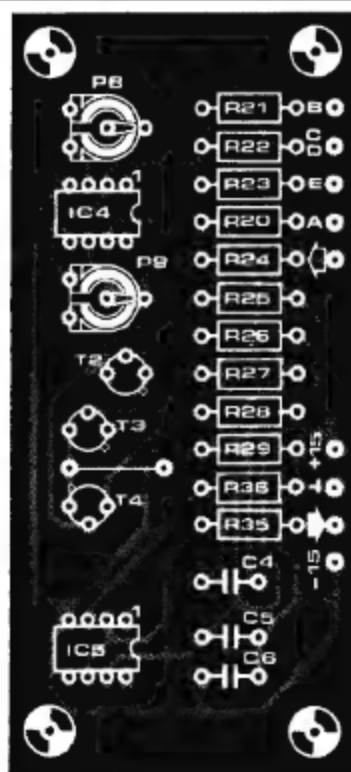
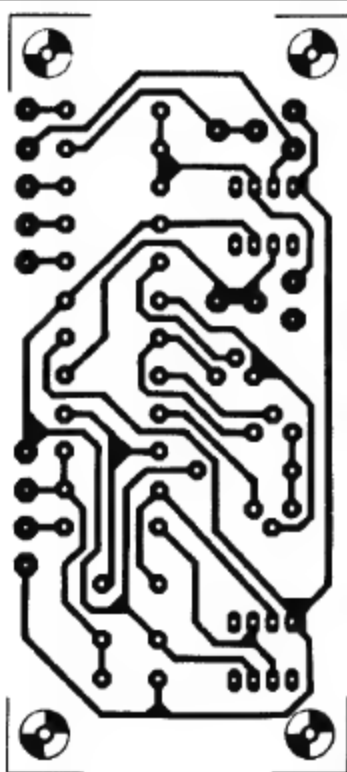
Figures

- Figure 1a. Control voltage and oscillator section of the LF-VCO module.
- Figure 1b. Sawtooth converter for a falling-edge sawtooth wave.
- Figure 1c. The pulse-width modulator generates a square wave from a sawtooth signal; the pulse/pause ratio can be modulated by a voltage at the PWM input.
- Figure 2. Layout, component placement plan, and parts list for the LF-VCO module. R14 is soldered onto the track side.
- Figure 3. LF-VCO front panel.
- Figure 4. Layout and component placement plan for the PWM circuit. The components are listed in the parts list for Figure 2.
- Figure 5. Assembly diagram for the LF-VCO, PWM board, and front panel.
- Figure 6. Wiring diagram for the front panel components of the pulse-width modulator.



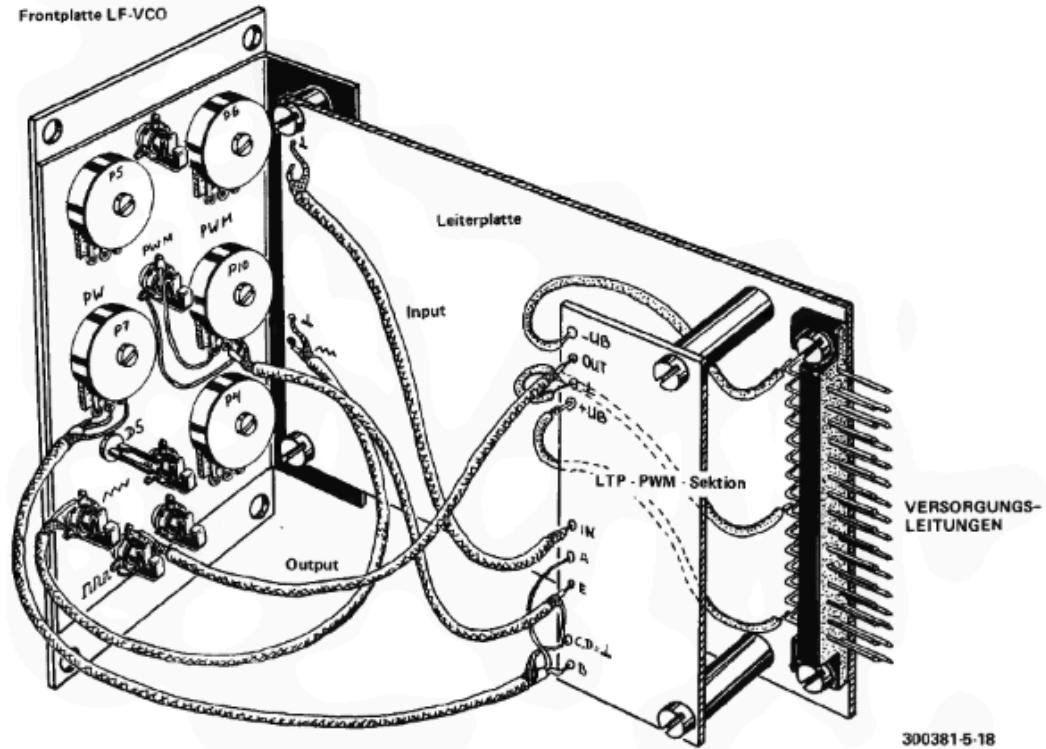


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Frontplatte L.F.-VCO



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